

Mohammad Atif Quamar

MSc in Machine Learning at MBZUAI

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Research Interests

Reinforcement Learning, Efficient Training, AI Alignment and Safety, Multi-modal Reasoning, Multi-Agentic Systems

Education



M.Sc. in Machine Learning 2026 – 2028

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE

• Relevant Coursework: Machine Learning, Mathematical Foundations of Machine Learning, Principles of Applied Statistics, Deep Learning, Machine Learning with Python



B.Tech. in Computer Science and Biosciences 2020 – 2024

Indraprastha Institute of Information Technology (IIIT), Delhi, India

• Relevant Coursework: Machine Learning, Computer Vision, Natural Language Processing, Artificial Intelligence, Linear Algebra, Multivariate Calculus, Differential Equations

Publications

* *Equal contribution*

- [P1] **Adaptive Blockwise Search: Inference-Time Alignment for Large Language Models**
Mohammad Atif Quamar^{*}, Mohammad Areeb^{*}, Nishant Sharma, Ananth Shreekumar, Jonathan Rosenthal, Muslum Ozgur Ozmen, Mikhail Kuznetsov and Z. Berkay Celik
arXiv, 2025. Under Review. [\[arXiv\]](#)
- [P2] **Reliable Chain-of-Thought via Prefix Consistency**
Naoto Iwase, Yuki Ichihara, Mohammad Atif Quamar and Junpei Komiyama
arXiv, 2026. Under Review. [\[arXiv\]](#)
- [P3] **Privileged Solutions or Context-Induced Teacher Behaviour? Dissecting On-Policy Self Distillation**
Yuki Ichihara, Naoto Iwase, Mohammad Atif Quamar and Junpei Komiyama
arXiv, 2026. Under Review. [\[arXiv\]](#)
- [P4] **Prefix-Denoising Consistency for Diffusion Language Models**
Yuki Ichihara, Naoto Iwase, Mohammad Atif Quamar, Junpei Komiyama
arXiv, 2026. Under Review. [\[arXiv\]](#)
- [W1] **STARS: Synchronous Token Alignment for Robust Supervision in LLMs**
Mohammad Atif Quamar^{*}, Mohammad Areeb^{*}, Mikhail Kuznetsov, Muslum Ozgur Ozmen, and Z. Berkay Celik
Structured Probabilistic Inference & Generative Modeling Workshop at ICML 2026. [\[arXiv\]](#)
- [W2] **Logit-Entropy Adaptive Stopping Heuristic for Efficient Chain-of-Thought Reasoning**
Mohammad Atif Quamar and Mohammad Areeb
Efficient Reasoning Workshop at NeurIPS 2025. [\[arXiv\]](#) [\[OpenReview\]](#)

[W3]

Decoding Histone Modification Signatures of Non-Coding RNAs via FMs

Nishant Sharma, **Mohammad Atif Quamar** and Pengtao Xie

Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences at NeurIPS 2025. [\[OpenReview\]](#)

Research Experience



Visiting Researcher

Feb 2026 – Jul 2026

MBZUAI, Abu Dhabi, UAE

Advisor: [Prof. Junpei Komiyama](#)

- Working on methods to improve reasoning in language models through reinforcement learning, self-consistency, and test-time inference strategies, to enhance mathematical reasoning and coding performance while reducing unreliable reasoning paths. Publication - [\[P2\]](#).



Research Intern

Sep 2025 – Feb 2026

University of Virginia, Virginia, United States

Advisor: [Prof. Ferdinando Fioretto](#)

- Developed a drift-resilient memory framework for code-execution agents using KL-constrained adapter updates, mitigating embedding distribution shifts during online learning to reduce unsafe code generation without compromising task success rates.



Research Intern

Aug 2025 – Sep 2025

UC San Diego, California, United States

Advisor: [Prof. Biwei Huang](#)

- Proposed a multimodal Chain-of-Thought framework that interleaves text with latent visual representations in VLMs, with a two-stage SFT + RL training setup for modality switching, achieving strong gains across 11 multimodal reasoning benchmarks. Publication - [\[P3\]](#).



Research Intern

Mar 2025 – Jul 2025

Purdue University, Indiana, United States

Advisor: [Prof. Z. Berkay Celik](#)

- Studied Bayesian sampling and inference-time alignment methods for language models, improving response quality and steering outputs toward harmlessness and positive sentiment. Publications - [\[C1\]](#) and [\[P1\]](#).

Industry Experience

Co-Founder

Sep 2023 – Jan 2025

Insituate, New Delhi, India

- Founded Insituate and grew it to \$250K ARR, building on-premise AI agents using open-source LLMs and RAG systems for enterprise use cases; deployments included Singapore Judiciary, Indian High Courts, Mizuho Bank, and PNC Bank.

Awards & Achievements

2026	MBZUAI Scholarship. 100% tuition scholarship for M.Sc. in ML; \$100K scholarship package.
2024	NASSCOM GenAI Cohort. Selected among the top 15 AI startups in India.
2024	NVIDIA Inception. Secured \$150K worth of AWS credits.
2024	Microsoft Founder's Hub. Received \$150K worth of Azure credits.
2020	JEE Main. Ranked in the top 3% among 1.1 million candidates.
Present	Competitive Programming. 1800+ rated on CodeChef and LeetCode; world top 3%.

Academic Service

2025	Program Committee. AAAI AI Alignment Track.
2025	Program Committee. Efficient Reasoning Workshop at NeurIPS.
2024	Teaching Assistant. ECO332: Valuation and Portfolio Management at IIITD.
2023	Teaching Assistant. DES102: Introduction to HCI at IIITD.